Computational Physics Lectures

Week 6: Free Energy and Potential of Mean Force

Bjørk Hammer
Department of Physics and Astronomy
Aarhus University
Denmark

Contents

6 Free energy and potential of mean force **3**

6.1 Helmholtz free energy 3

6.2 Potential of mean force 5

6.3 Harmonic integration and entropic stabilization 6

6.4 A reduced free-energy variational principle 7

6.5 Systems with well-separated configurational basins 8

6.6 Summary 9

© Bjørk Hammer

September 21, 2026

Week 6

Free energy and potential of mean force

Abstract. This chapter introduces the Helmholtz free energy and shows that canonical equilibrium minimizes a free-energy functional at fixed temperature. Integrating out microscopic coordinates leads to the potential of mean force, which governs the equilibrium distribution of the retained coordinates. Harmonic approximations are then used to connect energetic and entropic contributions to the equilibrium populations of configurational basins.

6.1 Helmholtz free energy

For an isolated compound system, equilibrium maximizes the total entropy at fixed total energy. The observed system alone, however, is in contact with a heat bath, so its energy is not fixed. If the observed system gains a small amount of energy δE, the bath loses the same amount and its entropy changes by approximately $-\delta E/T$. At fixed bath temperature, equilibrium must therefore reflect changes in both the energy and the entropy of the observed system. We now derive the thermodynamic quantity that combines these two contributions and subsequently show that it is minimized at canonical equilibrium.

Let the equilibrium phase-space distribution of a classical system be

$$P_{\mathrm{eq}}(\mathbf{q},\mathbf{p}) = \frac{1}{Z}\exp[-\beta H(\mathbf{q},\mathbf{p})], \qquad \beta = \frac{1}{k_{\mathrm{B}}T}, \tag{6.1}$$

where

$$Z = \frac{1}{h^{N_{\mathrm{s}}}}\int \exp[-\beta H(\mathbf{q},\mathbf{p})]\,\mathrm{d}^{N_{\mathrm{s}}}q\,\mathrm{d}^{N_{\mathrm{s}}}p. \tag{6.2}$$

Inserting the equilibrium distribution into the Gibbs entropy gives

$$\begin{aligned} S_{\mathrm{eq}} &= -\frac{k_{\mathrm{B}}}{h^{N_{\mathrm{s}}}}\int P_{\mathrm{eq}}\ln P_{\mathrm{eq}}\,\mathrm{d}^{N_{\mathrm{s}}}q\,\mathrm{d}^{N_{\mathrm{s}}}p \\ &= \frac{U_{\mathrm{eq}}}{T} + k_{\mathrm{B}}\ln Z, \end{aligned} \tag{6.3}$$

where $U_{\mathrm{eq}} = \langle H\rangle_{\mathrm{eq}}$. Rearranging gives

$$\boxed{F = U_{\mathrm{eq}} - TS_{\mathrm{eq}} = -k_{\mathrm{B}}T\ln Z.} \tag{6.4}$$

Here F denotes the equilibrium Helmholtz free energy.

Two paths to the equilibrium distribution

We now have two possible paths for determining the equilibrium distribution of a system in contact with a heat bath. They describe the same physical equilibrium, but they begin from different quantities being prescribed.

In the first path, the mean energy U of the system is prescribed. We then maximize the entropy $S[P]$ over all normalized trial distributions P that satisfy

$$\langle H\rangle_P = U. \tag{6.5}$$

The Lagrange multiplier associated with this energy constraint is identified with $1/T$. This is the maximum-entropy approach developed in the preceding chapter.

In the second path, the temperature T of the heat bath is prescribed instead. The mean energy is then not imposed as a constraint, because the system can exchange energy with the bath. As the trial distribution P is varied, both $\langle H\rangle_P$ and $S[P]$ may therefore change. We combine them in the free-energy functional

$$\mathcal{F}_T[P] = \langle H\rangle_P - TS[P]. \tag{6.6}$$

At fixed T, we minimize $\mathcal{F}_T[P]$ subject only to the normalization of P. This is the minimum-free-energy approach.

The two paths can therefore be summarized as follows:

$$\begin{aligned} &\text{prescribed } U: && \text{maximize } S[P] \text{ subject to normalization and } \langle H\rangle_P = U,\\ &\text{prescribed } T: && \text{minimize } \mathcal{F}_T[P] \text{ subject to normalization.} \end{aligned} \tag{6.7}$$

Although the constraints are different, both paths lead to the same Boltzmann distribution. We now show why the equilibrium distribution minimizes the free-energy functional when T is prescribed.

What is the Kullback–Leibler divergence?

Let P be any normalized trial distribution and let P_{eq} be the Boltzmann equilibrium distribution at the temperature imposed by the bath. The Kullback–Leibler divergence is defined by

$$D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}) = \frac{1}{h^{N_{\mathrm{s}}}}\int P\ln\!\left(\frac{P}{P_{\mathrm{eq}}}\right)\mathrm{d}^{N_{\mathrm{s}}}q\,\mathrm{d}^{N_{\mathrm{s}}}p. \tag{6.8}$$

It provides a measure of how much the trial distribution P differs from the equilibrium distribution P_{eq}. It is not a distance in the strict mathematical sense, because it is generally not symmetric:

$$D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}) \neq D_{\mathrm{KL}}(P_{\mathrm{eq}}\|P). \tag{6.9}$$

The two properties needed here are

$$D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}) \geq 0, \tag{6.10}$$

and

$$D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}) = 0 \quad\Longleftrightarrow\quad P = P_{\mathrm{eq}} \tag{6.11}$$

(up to sets of zero probability). Thus the divergence is positive for every trial distribution different from equilibrium and vanishes only for the equilibrium distribution itself.

We can now use these properties to establish the minimum principle. For the Boltzmann distribution,

$$P_{\mathrm{eq}} = \frac{1}{Z}\exp(-\beta H), \qquad \ln P_{\mathrm{eq}} = -\beta H - \ln Z. \tag{6.12}$$

Inserting the latter expression into the definition of the KL divergence and using the Gibbs entropy gives

$$\begin{aligned} k_{\mathrm{B}}T D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}) &= \langle H\rangle_P - TS[P] + k_{\mathrm{B}}T\ln Z \\ &= \mathcal{F}_T[P] - F, \end{aligned} \tag{6.13}$$

where

$$F = -k_{\mathrm{B}}T\ln Z \tag{6.14}$$

is the equilibrium Helmholtz free energy. Rearranging gives

$$\boxed{\mathcal{F}_T[P] = F + k_{\mathrm{B}}T D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}).} \tag{6.15}$$

Because the KL divergence is non-negative,

$$\boxed{\mathcal{F}_T[P] \geq F.} \tag{6.16}$$

Equality holds only when $P = P_{\mathrm{eq}}$. Therefore the Boltzmann distribution minimizes the free-energy functional, and its minimum value is the equilibrium Helmholtz free energy F.

6.2 Potential of mean force

In many systems we are interested in only a few coordinates. Let x denote a retained coordinate and y collectively denote coordinates that will be eliminated. We write the configurational potential energy as $U(x, y)$. Momenta are assumed to have been integrated out; for standard Cartesian kinetic energy the momentum integral is independent of x and y and contributes only a constant to the free energy.

The joint configurational equilibrium density is proportional to

$$P_{\mathrm{eq}}(x, y) \propto \exp[-\beta U(x, y)]. \tag{6.17}$$

The marginal equilibrium density of x is therefore

$$P_{\mathrm{eq}}(x) = \frac{1}{Z_x}\int \exp[-\beta U(x, y)]\,\mathrm{d}y, \tag{6.18}$$

where Z_x is the necessary normalization factor. This motivates defining the *potential of mean force* $A(x)$ through

$$\boxed{\exp[-\beta A(x)] = \frac{1}{C_y}\int \exp[-\beta U(x, y)]\,\mathrm{d}y,} \tag{6.19}$$

where the constant C_y supplies the units needed to make the logarithm dimensionless. Equivalently,

$$A(x) = -k_{\mathrm{B}}T\ln\left[\frac{1}{C_y}\int \exp[-\beta U(x, y)]\,\mathrm{d}y\right]. \tag{6.20}$$

Changing C_y adds a constant to $A(x)$. The zero of a potential of mean force is therefore arbitrary, just like the zero of an ordinary potential energy. Neither normalized probabilities, free-energy differences, nor mean forces are affected by this constant.

The marginal distribution becomes

$$\boxed{P_{\mathrm{eq}}(x) = \frac{1}{Z_x}\exp[-\beta A(x)],} \qquad Z_x = \int \exp[-\beta A(x)]\,\mathrm{d}x. \tag{6.21}$$

Thus $A(x)$ is the effective free-energy landscape governing the equilibrium statistics of the retained coordinate. For equilibrium questions involving only x, this reduction is exact: it gives the correct marginal distribution and configurational partition function, up to the reference constants. The function $A(x)$ alone does not retain the conditional distribution of y, observables depending explicitly on y, or in general the dynamical memory and friction produced by eliminating y.

Why it is called a potential of mean force

For Cartesian coordinates, an x-independent integration domain, and an x-independent measure, differentiation gives:

$$\begin{aligned}
\frac{\mathrm{d}A}{\mathrm{d}x} &= -k_{\mathrm{B}}T\frac{\mathrm{d}}{\mathrm{d}x}\ln\left[\frac{1}{C_y}\int \exp[-\beta U(x,y)]\,\mathrm{d}y\right] \\
&= -k_{\mathrm{B}}T\frac{\mathrm{d}}{\mathrm{d}x}\ln\left[\int \exp[-\beta U(x,y)]\,\mathrm{d}y\right] \\
&= \frac{\int (\partial U/\partial x)\exp[-\beta U(x,y)]\,\mathrm{d}y}{\int \exp[-\beta U(x,y)]\,\mathrm{d}y} \\
&= \left\langle \frac{\partial U}{\partial x}\right\rangle_{y|x},
\end{aligned} \tag{6.22}$$

where C_y which is independent of x contributes only a constant inside the logarithm and disappears upon differentiation. Consequently,

$$\boxed{-\frac{\mathrm{d}A}{\mathrm{d}x} = \left\langle -\frac{\partial U}{\partial x}\right\rangle_{y|x}.} \tag{6.23}$$

The force derived from $A(x)$ is the microscopic force averaged over the conditional equilibrium distribution of all eliminated coordinates at fixed x. For nonlinear collective coordinates, additional Jacobian or metric terms may occur.

6.3 Harmonic integration and entropic stabilization

Consider

$$U(x,y) = V(x) + \frac{1}{2}k(x)[y - y_0(x)]^2, \qquad k(x) > 0. \tag{6.24}$$

Integrating over y gives

$$\int_{-\infty}^{\infty} \exp[-\beta U(x,y)]\,\mathrm{d}y = \exp[-\beta V(x)] \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\beta k(x)(y-y_0(x))^2\right] \mathrm{d}y$$
$$= \exp[-\beta V(x)] \sqrt{\frac{2\pi}{\beta k(x)}}. \tag{6.25}$$

The potential of mean force is therefore

$$\boxed{A(x) = V(x) + \frac{1}{2} k_{\mathrm{B}} T \ln k(x) + \text{constant}.} \tag{6.26}$$

Here and below, the additive constant includes the fixed reference measure needed to make the logarithm dimensionless. The function $y_0(x)$ shifts the center of the conditional Gaussian but does not change its width or integrated statistical weight, so it does not appear in $A(x)$. By contrast, a smaller $k(x)$ permits a broader range of y values. That larger conditional entropy lowers the free energy. This is a simple example of *entropic stabilization*: two values of x may have the same potential energy $V(x)$ but different free energies because the eliminated coordinate has different amounts of available phase space.

6.4 A reduced free-energy variational principle

Strictly, a fixed reference density $m(x)$ is needed to make the logarithm in the continuous entropy dimensionless. For the Cartesian coordinate considered here, we assume that $m(x)$ is constant and absorb the resulting additive constant into the entropy and free energy. We therefore omit the reference density from the notation and write

$$S_x[P] = -k_{\mathrm{B}} \int P(x) \ln P(x)\,\mathrm{d}x. \tag{6.27}$$

The reduced functional is

$$\mathcal{F}_x[P] = \int P(x) A(x)\,\mathrm{d}x - T S_x[P]. \tag{6.28}$$

Because $P_{\mathrm{eq}}(x) = Z_x^{-1} \exp[-\beta A(x)]$, the same KL argument as for the full canonical distribution gives

$$\boxed{\mathcal{F}_x[P] = F_x + k_{\mathrm{B}} T D_{\mathrm{KL}}(P \| P_{\mathrm{eq}}),} \tag{6.29}$$

where

$$F_x = -k_{\mathrm{B}} T \ln Z_x \tag{6.30}$$

and

$$D_{\mathrm{KL}}(P\|P_{\mathrm{eq}}) = \int P(x) \ln\left(\frac{P(x)}{P_{\mathrm{eq}}(x)}\right) \mathrm{d}x. \tag{6.31}$$

Hence $\mathcal{F}_x[P] \geq F_x$, with equality only for the equilibrium marginal. Notice that $\langle A \rangle_P$ is not held fixed. At prescribed bath temperature, the entire functional is minimized subject only to normalization.

6.5 Systems with well-separated configurational basins

Suppose $A(x)$ has minima associated with configurational basins $\mathcal{B}_1, \mathcal{B}_2, \ldots$. Define the contribution of basin i to the partition function and its basin free energy by

$$Z_i = \int_{\mathcal{B}_i} \exp[-\beta A(x)]\, \mathrm{d}x, \qquad F_i = -k_{\mathrm{B}} T \ln Z_i. \tag{6.32}$$

The basin population is

$$P_i = \frac{Z_i}{\sum_j Z_j} = \frac{\exp(-\beta F_i)}{\sum_j \exp(-\beta F_j)}, \tag{6.33}$$

and therefore

$$\boxed{\frac{P_j}{P_i} = \exp[-\beta(F_j - F_i)].} \tag{6.34}$$

This relation does not require a harmonic approximation. It requires only a meaningful definition of the basins whose populations are compared. Well-separated metastable wells make such a definition natural and relatively insensitive to the precise basin boundaries.

Harmonic approximation to one-dimensional basins

Near a minimum x_i, let

$$A(x) \simeq A(x_i) + \frac{1}{2}\kappa_i (x - x_i)^2, \qquad \kappa_i = A''(x_i) > 0. \tag{6.35}$$

If the Gaussian is narrow compared with the basin, then

$$Z_i \simeq \exp[-\beta A(x_i)] \sqrt{\frac{2\pi}{\beta \kappa_i}}. \tag{6.36}$$

For two wells,

$$\boxed{\frac{P_2}{P_1} \simeq \exp\{-\beta[A(x_2) - A(x_1)]\} \sqrt{\frac{\kappa_1}{\kappa_2}}.} \tag{6.37}$$

There are two distinct entropic contributions here. The value $A(x_i)$ already contains the entropy of coordinates eliminated when the PMF was constructed. The curvature factor accounts for the range of the remaining coordinate x inside basin i. Thus no entropy has been counted twice.

Harmonic approximation in two dimensions

The same result can be obtained directly from a two-dimensional potential. Near minimum (x_i, y_i), suppose

$$U(x, y) \simeq U_i + \frac{1}{2} k_{ix} (x - x_i)^2 + \frac{1}{2} k_{iy} (y - y_i)^2. \tag{6.38}$$

The local configurational partition function is

$$Z_i \simeq \frac{2\pi}{\beta \sqrt{k_{ix} k_{iy}}} \exp(-\beta U_i), \tag{6.39}$$

so the local free energy is

$$F_i = U_i + \frac{1}{2} k_{\mathrm{B}} T \ln(k_{ix} k_{iy}) + \text{constant}, \tag{6.40}$$

and

$$\frac{P_2}{P_1} = \frac{Z_2}{Z_1} \simeq \frac{\sqrt{k_{1x} k_{1y}}}{\sqrt{k_{2x} k_{2y}}} \exp[-\beta(U_2 - U_1)]. \tag{6.41}$$

A deeper minimum is energetically favoured, whereas a broader minimum is entropically favoured. Their combined effect is summarized by the basin free energy.

6.6 Summary

1. At fixed bath temperature, canonical equilibrium minimizes the free-energy functional $\mathcal{F}_T[P] = \langle H \rangle_P - TS[P]$.
2. The excess free-energy functional is $k_{\mathrm{B}} T D_{\mathrm{KL}}(P \| P_{\mathrm{eq}})$ and is therefore non-negative.
3. Integrating out coordinates produces a potential of mean force that governs the exact equilibrium marginal of the retained coordinates.
4. A broader distribution of eliminated coordinates lowers the potential of mean force through its entropic contribution.
5. Basin populations are determined by basin free energies through $P_j / P_i = \exp[-\beta(F_j - F_i)]$.
6. In a harmonic approximation, local curvatures quantify the entropic contribution associated with the width of a basin.

Appendix

A1 Partition Function for Classical Harmonic Oscillator

Consider a classical harmonic potential in d dimensions:

$$U(\vec{x}) = U_0 + \sum_{i=1}^{d} \frac{1}{2} k_i x_i^2$$

We compute the canonical partition function:

$$Z = \int \exp\left(-\beta U(\vec{x})\right) d^d x$$

where $\beta = \frac{1}{k_B T}$.

A1.1 1D Case

Potential energy:

$$U(x) = U_0 + \frac{1}{2} k x^2$$

Partition function:

$$\begin{aligned} Z &= \int_{-\infty}^{\infty} \exp\left(-\beta \left(U_0 + \frac{1}{2} k x^2\right)\right) dx \\ &= e^{-\beta U_0} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2} \beta k x^2\right) dx \\ &= e^{-\beta U_0} \sqrt{\frac{2\pi}{\beta k}} \end{aligned}$$

$$\boxed{Z_{\text{1D}} = \sqrt{\frac{2\pi k_B T}{k}} \cdot e^{-U_0/(k_B T)}}$$

A1.2 2D Case

Potential energy:

$$U(x, y) = U_0 + \frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2$$

Partition function:

$$\begin{aligned} Z &= \iint \exp\left(-\beta\left(U_0 + \frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2\right)\right) dx\, dy \\ &= e^{-\beta U_0}\left(\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\beta k_x x^2\right) dx\right)\left(\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\beta k_y y^2\right) dy\right) \\ &= e^{-\beta U_0} \cdot \sqrt{\frac{2\pi}{\beta k_x}} \cdot \sqrt{\frac{2\pi}{\beta k_y}} \\ &= \frac{2\pi}{\beta\sqrt{k_x k_y}} \cdot e^{-\beta U_0} \end{aligned}$$

$$\boxed{Z_{\mathrm{2D}} = \frac{2\pi k_B T}{\sqrt{k_x k_y}} \cdot e^{-U_0/(k_B T)}}$$